

Gate

An analogue-modelled gate, expander and ducker with a filtered key input, lookahead and MIDI triggering.



Overview

Gate is three tools in one rack unit. As a GATE it shuts the signal down by a set amount whenever it falls below the threshold - for tightening drums and cleaning up noise between phrases. As an EXPANDER it turns quiet passages down gradually, at a ratio, which is gentler and more natural. Flip either one to DUCK and the curve turns upside down: the signal is pulled down whenever the key is loud, for ducking music under a voice or a bass under a kick.

The gain moves like an analogue VCA: opening is fast and then settles, and the release is a natural exponential fade. HOLD keeps the gate open for a moment after the level drops, and HYSTERESIS makes it close a few dB lower than it opens, so decaying notes don't make it chatter.

What the gate listens to (the key) can be the input itself or a separate signal on channels 3/4, and it can be filtered with a high-pass, a low-pass and a focus bell, so the gate only responds to the part of the sound you want. Held MIDI notes can also open the gate, for rhythmic chopping in time with a MIDI part.

Reach for it for

- Tightening kick, snare and toms, and cutting spill between hits
- Removing hiss, hum and amp noise between phrases
- Gentle downward expansion on vocals, dialogue and room mics

- Ducking music or a bass under a voice or kick (DUCK with EXT key)
- 80s gated reverb and rhythmic MIDI-triggered chopping

Controls

Presets and size (top left)

Display	Shows the preset number and name. While you move a control it shows that control's name and value for a moment. Click it for the preset list.
PRESETS	Opens the list of factory presets.
◀ / ▶	Step to the previous or next preset (wrapping round).
SIZE S / M / L	Panel size: small, medium or large. The panel also scales to fit the plugin window.
?	Opens a pop-up with the design credit, the GFX website address and the address of this manual. Click anywhere to close it.

Displays

Level history	A scrolling picture of the last few seconds, from 0 to -80 dB. The dark fill is the input level, the green OUT line the output, the amber GAIN line the gain the gate is applying. The red line is the threshold; the dashed red line under it is where the gate closes (THRESHOLD minus HYSTERESIS). A blue KEY line appears when the key differs from the input (EXT key, key filters or a non-linked stereo mode). The green bar along the bottom marks when the gate was OPEN (or DUCKED).
Transfer curve	The input-to-output curve. The bright branch is the opening curve, the dim one the closing curve (the gap between them is the hysteresis). The amber dot rides on it with the live level. With the EXT key, it shows the gain against the key level instead.

METERS	IN (left and right), GR (gain reduction of each side, or mid and side in M/S) and OUT (left and right).
GATE lights	Two green lights show when each detector is open (L+R in LINK mode, L and R in DUAL, M and S in M/S). MIDI lights while a MIDI note holds the gate, LSTN while LISTEN is on.

Mode buttons

MODEL: GATE / EXP	GATE: when the key is above the threshold the gate is fully open; below it, the signal drops by RANGE. EXP: a downward expander - below the threshold the gain falls at RATIO (at 1:2, 1 dB under the threshold gives 2 dB less output), limited to RANGE.
FUNCTION: NORM / DUCK	NORM: classic gating or expansion. DUCK: the same curve flipped - when the key goes above the threshold, the signal is pulled DOWN by RANGE.
KNEE: HARD / SOFT	HARD: a clean switch at the threshold. SOFT: on the GATE, the gate part-opens over the 12 dB below the threshold; on the EXP, the ratio eases in over 10 dB. Softer and less obvious.
STEREO: LINK / DUAL / M/S	LINK: one detector (the louder channel) drives both sides, so the image stays put. DUAL: independent left and right gates. M/S: independent mid and side gates (the key is split into mid and side too).
Latency readout	Shows ZERO LATENCY, or the lookahead time in ms when LOOKAHEAD is in use.

DYNAMICS

THRESHOLD	The level where the gate opens, -80 to 0 dB (-40 dB by default).
RANGE	How far the signal is turned down when the gate is shut (or ducked), 0 to 80 dB (60 dB by default). Small amounts (10 - 20 dB) keep some of the sound and are more natural.

RATIO	The expansion ratio, 1:1 to 1:10 (1:2 by default). Used only in EXP; in GATE mode the knob is greyed and reads GATE.
ATTACK	How fast the gate opens (or the duck pulls down), 0.01 to 100 ms; the time is roughly to 95% of the way.
HOLD	How long the gate stays open after the key drops below the close threshold, 0 to 2000 ms (50 ms by default).
RELEASE	How fast the gate closes (or the duck recovers), 5 to 4000 ms (150 ms by default), as a natural exponential fade.
HYSTERESIS	The gate opens at THRESHOLD and closes at THRESHOLD minus this amount, 0 to 20 dB (4 dB by default). Stops chatter on decaying notes.
LOOKAHEAD	0 to 10 ms. The audio is delayed so the gate can open just before a hit arrives, keeping the transient intact. 0 reads ZERO and adds no latency; any other value is reported to REAPER, which compensates for it.

KEY

INT / EXT	INT: the gate listens to its own input. EXT: it listens to the key input on channels 3/4 (see below).
MIDI	When on, held MIDI notes force the gate open (or the duck down). Channel 1 works on both sides, channel 2 on the left (or mid), channel 3 on the right (or side). MIDI is passed through to the next plugin.
LISTEN	Plays the filtered key instead of the output, so you can hear what the gate is listening to while you set the filters.
HPF (yellow cap)	A high-pass filter on the key only, 20 Hz (OFF) to 5 kHz. Raise it so low spill (e.g. kick in a snare mic) doesn't open the gate.
LPF	A low-pass filter on the key only, 100 Hz to 20 kHz (fully right = OFF). Lower it so cymbals and hats don't open a kick or tom gate.

FOCUS	The centre of a bell filter on the key, 100 Hz to 10 kHz. Greyed while F.GAIN is at 0.
F.GAIN	Boost or cut of the focus bell, -12 to +12 dB. Boost to make the gate more sensitive to the instrument's main frequency.
LINK	In DUAL and M/S: how far each detector is pulled towards the louder side, 0 to 100%. 100% behaves like linked; 0% is fully independent. In LINK mode it is greyed and reads FULL.

MASTER

MIX	Blends the dry signal back in, 0 to 100%. 100% is the full effect; lower settings make the gate's cut shallower.
OUTPUT	Output level, -24 to +12 dB.

Knobs: drag up or down (Shift for fine moves); double-click or Ctrl-click to reset a knob to its default. Values show while you hover or drag.

Presets

A preset sets the dynamics, the mode buttons, the key section, MIX and OUTPUT, and switches LISTEN off.

Init	Every control back to its default.
Kick Tight	Fast gate with 2 ms lookahead; the key is filtered to 30 - 400 Hz with a boost at 80 Hz.
Snare Crack	Very fast gate, 40 dB range and 1.5 ms lookahead; key focused on 2 kHz.
Tom Cleanup	Soft-knee gate with a longer hold and release for ringing toms; key focused around 120 Hz.
Hi-Hat Clean	Short, fast gate with the key high-passed at 4 kHz.
Vocal Breath	Soft 1:2 expander that takes 12 dB off breaths and room noise.

Guitar Noise	Soft-knee gate at -55 dB to remove amp hiss between phrases.
Bass DI	1:3 expander for a DI bass, with a 200 ms release.
Gentle Expander	A light 1:1.5 soft expander with up to 15 dB of reduction.
Room Mic Tame	Soft 1:2.5 expander to pull room mics down between hits.
Dialogue Clean	Soft expander for speech, key focused on the voice around 2.5 kHz.
Kick Duck (Ext Key)	Ducks the signal by 9 dB from a kick on the key input (low-passed at 200 Hz).
Voice-Over Duck (Ext Key)	A slow 1:4 expander-ducker that lowers music by up to 12 dB under a voice on the key input.
Gated Reverb 80s	Hard gate with a 250 ms hold and a fast 40 ms release: the classic chopped reverb tail.
MIDI Chop	Threshold at 0 dB and MIDI on: only held MIDI notes open the gate.
Lookahead Drum	Very fast gate with 5 ms lookahead so the transients stay whole.

Using the key input (EXT)

1. On the track with Gate, open ROUTE and set the track to 4 channels.
2. On the track you want to key from (e.g. the kick or the voice), add a send to Gate's track and set its destination to channels 3/4.
3. Click EXT. Use LISTEN to check you can hear the key, set the key filters, then switch LISTEN off.
4. Set THRESHOLD so the key crosses it; with DUCK, adjust RANGE for how far the signal dips.

Using MIDI triggering

1. Put Gate on the audio track you want to chop, and make sure it receives MIDI: either a MIDI item on the same track before Gate, or a MIDI send from another track.
2. Click MIDI in the KEY section. Notes on channel 1 open both sides; channels 2 and 3 open left and right (mid and side in M/S) separately.
3. Turn THRESHOLD to 0 dB if you want only the MIDI notes to open the gate, not the audio itself.

Tips

- Set HYSTERESIS to a few dB before reaching for a longer HOLD: it stops chatter without keeping the gate open too long.
- For natural results on vocals and rooms, use EXP with a soft knee and 10 - 20 dB of RANGE rather than a full gate.
- Use the key filters with LISTEN: when you can hear only the drum you want in the key, the gate will only open for that drum.
- A few ms of LOOKAHEAD keeps the attack of fast drums intact even with a slow ATTACK setting.

File: gfx-gate.jsfx